PRODUCT COUNT ANALYSIS

Sample Superstore Dataset

TASK 15 | DATA ANALYTICS INTERNSHIP

Task	Task 15 — Product Count Analysis
Organization	Veda Technologies
Dataset	Sample Superstore
Tool	Microsoft Excel
Deliverables	Product count summary and largest category

This report presents an Excel-based approach to counting products by category in the Sample Superstore dataset. It explains the use of COUNTIF and COUNTIFS, distinguishes record counts from unique-product counts, and describes how to identify the category with the largest number of unique products.

Results note: Actual category counts are left blank because the source dataset was not attached for calculation. Enter the verified values from your workbook before submitting this report.

1. Objective

The objective is to count products by category, practice category-based Excel functions, check unique products, and identify the category with the highest unique-product count.

2. Dataset Overview

The Sample Superstore dataset commonly contains order, product, category, sales, quantity, discount, and profit information. Depending on the version, useful columns may include Product ID, Product Name, and Category. Common categories are Furniture, Office Supplies, and Technology.

3. Tools Used

Microsoft Excel is used to inspect the dataset, create formulas, build a product count table, and optionally use PivotTables or charts to present the results.

4. Counting Approach

A product may appear on multiple sales rows. Counting rows by category gives the number of records or line items. Counting distinct Product IDs gives the number of unique products. Because the task hints ask to check unique products, the unique-product count should be the main result when Product ID is available.

5. Formulas Used

Metric	Example formula	Purpose
Record count	=COUNTIF(Orders!C:C,A2)	Counts rows matching a category.
Multiple conditions	=COUNTIFS(Orders!C:C,A2,Orders!D:D,"West")	Counts rows meeting both conditions.
Unique products (Excel 365)	=COUNTA(UNIQUE(FILTER(Orders!B$2:B$10000,(Orders!C$2:C$10000=A2)*(Orders!D$2:D$10000="West"))))	Counts distinct products in a category.
Largest category	=INDEX(A2:A4,MATCH(MAX(B2:B4),B2:B4,0))	Returns the category with the highest count.

These are examples. Replace the sheet name, column letters, and ranges with those used in your workbook. The unique-product formula assumes Product ID is in column B and Category is in column C.

6. Methodology

Step	Action
1	Open the Sample Superstore workbook and identify the Category and Product ID columns.
2	Create a separate worksheet named Product Summary.
3	List the categories in the summary table.
4	Use COUNTIF to count category records and a distinct-count method for unique products.
5	Optionally create a PivotTable to summarize the category data.
6	Compare unique-product counts and identify the highest value.
7	Validate formulas, exclude blank product identifiers, and check for ties.

7. Results Summary

Category	Record count	Unique products
Furniture	_____	_____

Office Supplies	_____	_____
Technology	_____	_____
Largest category by unique products	—	_____

Fill in this table from the actual workbook results. Do not report the largest category until the unique-product counts have been verified.

8. Optional Visualization

A clustered bar chart can compare unique-product counts across categories. Select the Category and Unique Product Count columns, choose Insert → Bar Chart → Clustered Bar, title it 'Unique Products by Category', and enable data labels. The chart is optional; the task requires the product count table and top category.

9. Key Learnings

This task practices COUNTIF and COUNTIFS, unique-value analysis, PivotTables, and clear presentation of category-level summaries. It highlights why counting sales records and counting unique products answer different questions.

10. Conclusion

The Product Count Analysis workflow summarizes product variety by category and identifies the category with the highest number of unique products. Choosing the correct counting method and validating formulas help make the results reliable.